



R V COLLEGE OF ENGINEERING
(An autonomous institution affiliated to VTU, Belgaum)
DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

IM233AI – WORK SYSTEMS DESIGN

UNIT II

Topic ID 3: Introduction to Methods Engineering & Operations Analysis.

Q No	Question	Marks	Difficulty Level	BT Level	COs
1	Explain the role of Methods Engineering in improving productivity and work design.	8	M	2	2
2	List and explain the steps in the methods engineering procedure.	8	M	2	2
3	Explain the relationship between methods engineering and automation. How does methods analysis support automation decisions?	8	M	4	2
4	Explain how methods engineering contributes to lean manufacturing and continuous improvement.	8	M	4	2
5	How does the "USA Principle" of automation apply to Methods Engineering?	8	M	3	2

Topic ID 4: Charting and Diagramming Techniques.

6	Explain how flow process charts are used in methods engineering. Provide an example of how they help improve a manual task.	8	M	3	2
7	Explain the difference between operation process charts and flow process charts. Provide one example of each.	8	M	4	2
8	Describe how block diagrams are used in methods engineering. What are their advantages in system visualization?	8	M	2	2
9	Discuss the role of traditional IE charting techniques in identifying non-value-added activities. Provide two examples.	8	M	4	2



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10	Explain the construction and use of an "Operation Process Chart."	8	M	2	2
11	Describe the "Flow Diagram" (Spatial) and its relationship to the Flow Process Chart.	8	M	3	2
12	Explain the use of "Two-Handed Process Chart" (Right-hand/Left-hand chart) in the application of Principles of Motion Economy.	8	M	3	2
13	What is a Man-Machine Chart (Multiple Activity Chart), and how is it used to calculate efficiency?	8	M	2	2
14	Discuss "Check Sheets" and "Pareto Analysis" as data collection tools in Methods Engineering.	8	M	2	2
15	Review any four types of basic Data Collection and Analysis techniques in Methods Engineering.	8	M	2	2
16	Explain and illustrate the three stages of Automation Migration Strategy.	8	M	3	2
17	Explain any four techniques of Methods Engineering.	8	M	2	2

Topic ID 5: Principles of Motion Economy

18	Discuss how principles of motion economy are applied in methods engineering. Provide two examples.	8	H	4	2
19	Define the principles of motion economy. How do they contribute to efficient work design?	8	M	2	2
20	Explain any six principles of motion economy related to the use of the human body and explain each briefly.	8	M	2	2
21	Explain how workplace layout affects motion economy. Suggest three layout improvements for a manual assembly station.	8	M	3	2
22	Identify and explain three principles of motion economy related to the arrangement of the workplace.	8	M	2	2
23	Discuss the principles of motion economy specifically related to the use of the two hands during a manual task.	8	M	3	2



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24	"The hands should be relieved of work elements that can be performed by other body parts." Explain this principle with respect to jigs, fixtures, and foot pedals.	8	M	3	2
25	Describe the concept of "Normal" and "Maximum" Reach Zones in workstation design.	8	M	2	2